

GED Mathematics Worksheet

Slope and Equations of Lines

Name: _____ Date: _____

Score: _____ / 50

Part A: Calculating Slope from Two Points (Questions 1-9)

Find the slope of the line passing through each pair of points. Use the formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

1. Find the slope between $(2, 3)$ and $(5, 9)$.
2. Find the slope between $(-1, 4)$ and $(3, 2)$.
3. Find the slope between $(0, 0)$ and $(4, 8)$.
4. Find the slope between $(-3, -2)$ and $(1, 6)$.
5. Find the slope between $(5, 1)$ and $(5, 7)$.
6. Find the slope between $(2, -3)$ and $(8, -3)$.
7. Find the slope between $(-4, 3)$ and $(2, -9)$.

8. Find the slope between $(0, 5)$ and $(-3, 2)$.

9. Find the slope between $(-2, -5)$ and $(4, 7)$.

Part B: Identifying Parallel and Perpendicular Lines (Questions 10-17)

For each pair of slopes, determine if the lines are parallel, perpendicular, or neither.

10. $m_1 = 3$ and $m_2 = 3$

11. $m_1 = 2$ and $m_2 = -\frac{1}{2}$

12. $m_1 = \frac{3}{4}$ and $m_2 = \frac{3}{4}$

13. $m_1 = 5$ and $m_2 = -5$

14. $m_1 = -\frac{2}{3}$ and $m_2 = \frac{3}{2}$

15. $m_1 = \frac{1}{4}$ and $m_2 = -4$

16. $m_1 = -1$ and $m_2 = 1$

17. $m_1 = \frac{5}{6}$ and $m_2 = \frac{6}{5}$

Part C: Finding Equations of Lines (Questions 18-27)

Find the equation of the line given the slope and a point, or two points. Use point-slope form: $y - y_1 = m(x - x_1)$

18. Find the equation of a line with slope $m = 2$ passing through $(1, 3)$.
19. Find the equation of a line with slope $m = -3$ passing through $(2, -5)$.
20. Find the equation of a line passing through $(1, 2)$ and $(3, 6)$.
21. Find the equation of a line with slope $m = \frac{1}{2}$ passing through $(0, 4)$.
22. Find the equation of a line passing through $(-2, 3)$ and $(4, 6)$.
23. Find the equation of a line with slope $m = -\frac{2}{3}$ passing through $(3, 1)$.
24. Find the equation of a line passing through $(0, 0)$ and $(5, 10)$.
25. Find the equation of a line with slope $m = 0$ passing through $(2, 5)$.
26. Find the equation of a line passing through $(-3, -1)$ and $(2, 4)$.
27. Find the equation of a line with slope $m = -1$ passing through $(4, -2)$.

Part D: Graphing Linear Equations (Questions 28-33)

Write the equation in slope-intercept form and identify the slope and y -intercept.

28. $2x + y = 5$

29. $3x - y = 6$

30. $4x + 2y = 8$

31. $y = 3x - 2$

32. $-2x + 3y = 9$

33. $5x - 5y = 10$

Part E: Real-World Applications of Slope (Questions 34-38)

34. A wheelchair ramp rises 2 feet over a horizontal distance of 24 feet. What is the slope of the ramp?
35. A staircase has steps that rise 7 inches for every 10 inches of horizontal distance. What is the slope of the staircase?
36. A road elevation increases 300 feet for every 2000 feet of horizontal distance. Express this as a slope in simplest form.

37. A mountain trail descends 400 feet in elevation over a horizontal distance of 1 mile (5280 feet). What is the slope of the trail?
38. A ladder leans against a wall, reaching 12 feet high. The base of the ladder is 5 feet from the wall. What is the slope of the ladder?

Part F: Distance Between Points (Questions 39-43)

Find the distance between each pair of points using the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

39. Find the distance between $(0, 0)$ and $(3, 4)$.
40. Find the distance between $(1, 2)$ and $(4, 6)$.
41. Find the distance between $(-2, 1)$ and $(3, 5)$.
42. Find the distance between $(0, -3)$ and $(4, 0)$.
43. Find the distance between $(-1, -1)$ and $(2, 3)$.

Part G: Systems of Equations (Questions 44-48)

Solve each system of equations using substitution or elimination.

44.
$$\begin{cases} y = 2x + 1 \\ y = -x + 4 \end{cases}$$

$$45. \begin{cases} x + y = 5 \\ x - y = 1 \end{cases}$$

$$46. \begin{cases} 2x + y = 7 \\ x - y = 2 \end{cases}$$

$$47. \begin{cases} 3x - 2y = 4 \\ x + y = 5 \end{cases}$$

$$48. \begin{cases} y = \frac{1}{2}x + 3 \\ y = -2x + 8 \end{cases}$$

Part H: Mixed Review (Questions 49-50)

49. Two parallel lines have equations $y = 3x + 2$ and $y = 3x - 5$. Find the perpendicular slope to these lines, and write an equation of a line that passes through $(2, 4)$ with that perpendicular slope.

50. A line passes through the points $(-2, 1)$ and $(4, 7)$. Find the slope, write the equation in slope-intercept form, and find the distance between the two points.